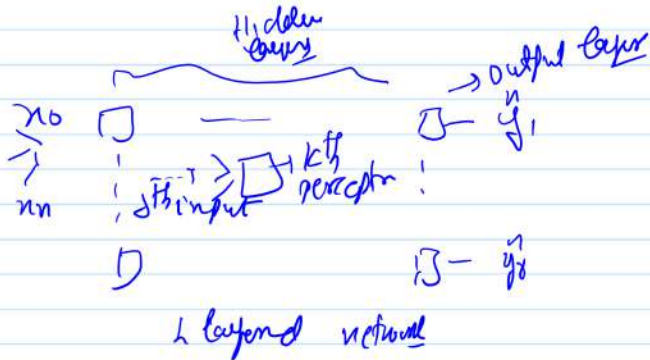


COL774 / COL7341

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$$J(\theta) = \frac{1}{2} \sum_{k=1}^m (y_k - \hat{y}_k)^2$$

Assume $m=2$
(single example)

$$\theta = \{\theta^{(l)}\}_{l=1}^L$$

$$\theta^{(l)} = n^{(l)} \begin{bmatrix} \theta_{k1}^{(l)} \\ \theta_{k2}^{(l)} \\ \vdots \\ \theta_{kL}^{(l)} \end{bmatrix}$$

$$O_k^{(l)} = g(\theta_k^{(l)T} O^{(l-1)})$$

$$\text{net}_k^{(l)} = \theta_k^{(l)T} O^{(l-1)} - (1)$$

$O_k^{(l)} :=$ k th output of l th layer

$$\nabla_{\theta} J(\theta)$$

parameters in network

To achieve objective

(A) $l=2$ (output layer)

$$\hat{y}_k = O_k^{(2)}$$

$$\frac{\partial J(\theta)}{\partial \text{net}_k^{(2)}} = \frac{\partial \left[\frac{1}{2} \sum_{k=1}^m (y_k - O_k^{(2)})^2 \right]}{\partial \text{net}_k^{(2)}}$$

$$= (y_k - O_k^{(2)})(-1) \cdot \frac{\partial O_k^{(2)}}{\partial \text{net}_k^{(2)}}$$

$$= (y_k - O_k^{(2)})(-1) O_k^{(2)} (1 - O_k^{(2)})$$

objective $\Rightarrow$ Compute $\frac{\partial J(\theta)}{\partial \text{net}_k^{(l)}} \neq 0, k$
then express $\frac{\partial J(\theta)}{\partial \text{net}_k^{(l)}}$ in terms of $\frac{\partial J(\theta)}{\partial \text{net}_k^{(l-1)}}$

(B)

$$\frac{\partial J(\theta)}{\partial \text{net}_k^{(l)}} \quad l \neq L$$

$$\frac{\partial J(\theta)}{\partial O_k^{(l)}} \cdot \frac{\partial O_k^{(l)}}{\partial \text{net}_k^{(l)}} \rightarrow O_k^{(l)} (1 - O_k^{(l)})$$

$$\frac{\partial J(\theta)}{\partial O_k^{(l)}} = \sum_{s=1}^L \frac{\partial J(\theta)}{\partial \text{net}_s^{(l+1)}} \cdot \frac{\partial \text{net}_s^{(l+1)}}{\partial O_k^{(l)}} - (4)$$

Aside: $x \xrightarrow{f_1} z_1 \xrightarrow{f_2} y \rightarrow R \quad y = h(f_1(x), f_2(x), \dots, f_n(x))$

Let us define

$$\frac{\partial J(0)}{\partial \text{net}_k^{(l)}} = -\delta_k^{(l)} \quad \left| \quad \begin{array}{l} \frac{\partial y}{\partial a} = \sum_{s=1}^p \frac{\partial y}{\partial z_s} \cdot \frac{\partial z_s}{\partial a} \quad n \rightarrow z \rightarrow y \\ z \rightarrow (2) \end{array} \right.$$

$$\frac{\partial \text{net}_s^{(l+1)}}{\partial \delta_k^{(l)}} = \theta_{sk}^{(l+1)} \quad \text{--- (5)}$$

$$\begin{array}{l} z \rightarrow \varepsilon_{p^1} \rightarrow z \rightarrow \varepsilon_{p^2} \rightarrow y \\ \nabla_z y = \left(\frac{\partial z}{\partial x} \right)^T \nabla_x y \quad \leftarrow \text{--- (3)} \\ \left(\frac{\partial z}{\partial x} \right) = \begin{bmatrix} \uparrow & \downarrow \\ \frac{\partial z}{\partial x} & -\frac{\partial z_s}{\partial x} \end{bmatrix} \end{array}$$

↳ Jacobian

Summary: -

(A) $\frac{\partial J(0)}{\partial \text{net}_k^{(2)}} = (y_k - o_k^{(2)}) (-v) \quad (o_k^{(2)} (1 - o_k^{(2)})) \quad \text{--- (7)}$

(B) $\frac{\partial J(0)}{\partial \text{net}_k^{(e)}} = \left(\frac{\partial J(0)}{\partial o_k^{(e)}} \right)_{T1} \cdot \left(\frac{\partial o_k^{(e)}}{\partial \text{net}_k^{(e)}} \right)_{T2}$

$$= \left[\sum_{s=1}^p \frac{\partial J(0)}{\partial \text{net}_s^{(l+1)}} \cdot \frac{\partial \text{net}_s^{(l+1)}}{\partial \delta_k^{(l)}} \right] \cdot o_k^{(e)} (1 - o_k^{(e)})$$

$$= \left[\sum_{s=1}^{n^{(l+1)}} -\delta_s^{(l+1)} \cdot \theta_{sk}^{(l+1)} \right] o_k^{(e)} (1 - o_k^{(e)}) \quad \text{--- (8)}$$

Expressing

$\frac{\partial J(0)}{\partial o_{kj}^{(p)}}$ in terms of $\frac{\partial J(0)}{\partial \text{net}_k^{(p)}}$

$$\frac{\partial J(0)}{\partial o_{kj}^{(p)}} \rightarrow \frac{\partial J(0)}{\partial \text{net}_k^{(p)}} \cdot \left[\frac{\partial \text{net}_k^{(p)}}{\partial o_{kj}^{(p)}} \right] \Rightarrow o_{kj}^{(p-1)}$$

$\downarrow$
 $-\delta_k^{(p)}$

Final Backpropagation Algorithm: -

For $l = L$ to 1

if $l = L$

compute $\frac{\partial J(0)}{\partial \text{net}_k^{(L)}}$ directly $\forall k$
(Equation 7)

pre

compute $\frac{\partial J(\theta)}{\partial w_k}$ in terms
of $\left\{ \frac{\partial J(\theta)}{\partial w_s^{(t+1)}} \right\}_{s=1}^{n^{(t+1)}}$ in next layer
(Equation 8)

3 vl:-

→ Final update:-

$$\theta^{(t+1)}(t+1) \leftarrow \theta^{(t+1)}(t) - \gamma \cdot \nabla J(\theta)_{\theta^{(t+1)}(t)}$$